

Proofs

Formal Logic

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Direct Democracy

Contrapositive: every fair democracy has a tiebreaking rule.

Suppose that a country does not have a tiebreaking rule. Then, they either have elections without tiebreaks, or no elections at all.

If no elections, then clearly not a democracy.

If no tiebreaks, then someone must have outsized influence over the outcome (be decisive) – but then, they cannot be a democracy.

Median Voter Theorem

Proof by contradiction: suppose not. Then, there exists some equilibrium whereby either A or B is playing a policy separate from the median voter's preferences.

Then, either A is closer to M than B , or B is closer to M than A , or A and B are some equal (non-zero) distance away from M .

If one is further than the other, the closer one will win – so, the further one cannot be playing an optimal point.

If they are the same (non-zero) distance away, either one could win by moving just a little bit closer to M – so, neither of these players can be playing an optimal point.

Therefore, it cannot be the case that $A \neq M$ or $B \neq M$ in equilibrium. Therefore, $A = B = M$ in equilibrium.